

TECHNICAL NARRATIVE REPORT

FINAL TILANGGIT

An Impact Assessment Comparing Program Beneficiaries and Non-Beneficiaries

Bureau of Fisheries and Aquatic Resources

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EXECUTIVE SUMMARY

This technical report presents a socioeconomic impact evaluation of the Final Tilanggit program, a livelihood intervention implemented by the Philippine Bureau of Fisheries and Aquatic Resources. The evaluation aims to assess the socioeconomic conditions of program beneficiaries compared to non-beneficiaries and to estimate the program association with participant outcomes. The evaluation design is a quasi-experimental cross-sectional comparison of program beneficiaries and non-beneficiaries using a no-baseline framework with indicator comparisons of beneficiary versus non-beneficiary distributions.

The evaluation surveyed 300 respondents in the Beneficiary group and 300 respondents in the Non-Beneficiary group using distinct questionnaires. To estimate the impact on total current income coded levels, propensity score matching was conducted using a gradient boosting propensity model scored out-of-fold across 5 folds, followed by 1-to-1 nearest-neighbour matching on the logit propensity score within a caliper of 0.2 standard deviations. Covariate balance was checked using standardized mean differences against a threshold of 0.1, with 297 matched pairs formed out of 600 respondents analysed.

The average treatment effect on the treated estimated using the matched pairs was 0.205, with a 95% confidence interval of [0.074, 0.35] and a paired-test p-value of 0.004. However, balance was not fully achieved as the mean absolute standardized mean difference after matching was 0.126, with 16 features exceeding the threshold against an allowed maximum of 8. Therefore, the estimated association must be interpreted cautiously.

In conclusion, while the matching analysis shows a statistically significant positive association between program participation and the coded level of total current income, the lack of complete covariate balance and the cross-sectional nature of the data mean that causal claims cannot be definitively established beyond the matching assumptions.

Major policy recommendations advanced by this report:

- Expand the distribution of essential processing and packaging equipment, such as vacuum packers and solar dryers, to address the remaining gaps where substantial percentages of beneficiaries still report lacking equipment.
- Strengthen support for organized fisherfolk associations, given that membership strongly predicts program participation and correlates with active livelihood engagement.
- Enhance technical training programs focusing on fish processing methods to further increase the proportion of beneficiaries actively engaging in value-adding activities like drying and deboning.
- Target outreach to non-beneficiary communities to increase general program awareness and extend livelihood support to eligible households currently outside the program.
- Incorporate baseline data collection in future iterations of program evaluations to strengthen causal inference and bypass the limitations of cross-sectional matching designs.

1.0 INTRODUCTION

1.1 RATIONALE OF THE PROJECT

The Philippine Bureau of Fisheries and Aquatic Resources implements livelihood programs such as the Final Tilanggit to support fisherfolk communities, improve processing capabilities, and enhance overall socioeconomic well-being. Evaluating these interventions is crucial for understanding how resource allocation and asset distribution affect recipient households and whether program objectives are being met in terms of tangible asset acquisition and income improvement.

This evaluation matters because empirical evidence on program effectiveness informs policy adjustments and future project designs. By examining specific indicators related to equipment ownership, processing activities, and livelihood perceptions, this report provides technical insights into the differentials between program beneficiaries and non-beneficiaries, assisting administrators in refining support mechanisms for coastal populations.

1.2 OBJECTIVES OF THE PROJECT

To evaluate the socioeconomic impact of the Final Tilanggit program on fisherfolk beneficiaries by comparing their outcomes and estimating the program association relative to non-beneficiaries. Specifically, it aimed to:

- To compare the distribution of equipment ownership, processing methods, and livelihood activities between beneficiaries and non-beneficiaries.
- To evaluate differences in programmatic awareness and participation in fisherfolk associations between the two groups.
- To assess subjective ratings regarding livelihood confidence, future expectations, and community participation.
- To estimate the average treatment effect of program participation on coded total current income using propensity score matching

2.0 METHODOLOGY

2.1 RESEARCH DESIGN

The research design employed a quasi-experimental cross-sectional comparison of program beneficiaries and non-beneficiaries under a no-baseline framework. Indicator comparisons utilized beneficiary versus non-beneficiary distributions from surveyed respondents.

The study engaged a total of 600 respondents, divided equally into 300 respondents in the Beneficiary group using the Tilanggit Survey – Benepisyaryo instrument and 300 respondents in the Non-Beneficiary group using the Tilanggit Survey – Hindi Benepisyaryo instrument.

2.2 SURVEY INSTRUMENTS AND RESPONDENTS

Beneficiary instrument	Tilanggit Survey – Benepisyaryo (140 survey questions)
Non-Beneficiary instrument	Tilanggit Survey – Hindi Benepisyaryo (116 survey questions)
Respondents	300 benchmark / 300 current responses from fisherfolk participants
Comparable indicators	102 paired items

2.3 STATISTICAL TREATMENT

Indicator comparisons between the Beneficiary and Non-Beneficiary groups were conducted by examining the percentage distributions of respondents choosing each response option across 102 compared totals and 60 listed indicators, ranked with the largest differences first.

Propensity score matching was executed using a gradient boosting propensity model scored out-of-fold across 5 folds. The analysis used 1-to-1 nearest-neighbour matching on the logit propensity score within a caliper of 0.2 standard deviations, utilizing 30 features used by the model out of a total set of features.

Covariate balance was evaluated using standardized mean differences against a threshold of 0.1. The average treatment effect on the treated was calculated along with its 95% confidence interval and paired-test p-value across 297 matched pairs.

Note: Comparisons are limited to questions present in both instruments; unmatched questions are excluded from paired analysis. Differences in respondent counts between phases may affect comparability and are flagged in the findings where relevant

3.0 RESEARCH FINDINGS

4.0 SYNTHESIS AND IMPLICATIONS

This section synthesizes the key findings drawn from the preceding analyses and highlights their implications for policy, program design, and community-level development.

4.1 SUMMARY OF FINDINGS

1. All 300 beneficiaries (100%) reported receiving support such as equipment, training, or capital from the Tilanggit Program, whereas all 300 non-beneficiaries (100%) reported receiving no such support.
2. Among beneficiaries, 85.3% reported actively using the knowledge or equipment provided, while 14.7% reported not using them, with 0% of non-beneficiaries responding affirmatively.
3. Packaging equipment ownership currently was reported as present by 50.3% of beneficiaries compared to 8.3% of non-beneficiaries, while production equipment scales were present for 70% of beneficiaries versus 29% of non-beneficiaries.

4. Fish processing activities differed notably, with 54.3% of beneficiaries engaged in drying (tilanggit/daing) compared to 22.7% of non-beneficiaries, and 26.7% reporting no processing versus 64.3% of non-beneficiaries.
5. Membership in a livelihood-related association or club was reported by 52.7% of beneficiaries and 27% of non-beneficiaries.
6. Household frequency of making or purchasing tilanggit in the past month was higher among beneficiaries, with 43.7% reporting 6 times or more compared to 7.7% of non-beneficiaries, while 7% of beneficiaries and 38.7% of non-beneficiaries reported none.
7. Subjective ratings on a 1 to 5 scale showed that beneficiaries reported higher mean scores for livelihood confidence for the next year (3.8 versus 3.3), access to support or services when needed (3.5 versus 3), active community participation (3.8 versus 3.3), meeting basic family needs (3.4 versus 3), confidence in joining a new program or business (3.9 versus 3.5), and community mutual aid (4 versus 3.8) compared to non-beneficiaries.
8. For total current income from all sources, 46% of beneficiaries and 48.3% of non-beneficiaries reported being in the lowest coded bracket of ₱10K or less, while higher brackets showed 37.3% of beneficiaries and 42.7% of non-beneficiaries in the >₱10K–₱20K range, 13% of beneficiaries and 6.3% of non-beneficiaries in the >₱20K–₱30K range, 3% of beneficiaries and 2% of non-beneficiaries in the >₱30K–₱40K range, 0.3% of beneficiaries and 0.7% of non-beneficiaries in the >₱40K–₱50K range, and 0.3% of beneficiaries and 0% of non-beneficiaries in the >₱50K range.
9. The propensity score matching analysis yielded an average treatment effect on the treated of 0.205 for the coded total current income outcome, supported by a 95% confidence interval of [0.074, 0.35] and a paired-test p-value of 0.004 across 297 matched pairs.

4.2 CONCLUDING REMARKS

The evaluation demonstrates that beneficiaries of the Final Tilanggit Program exhibit higher levels of asset ownership, processing engagement, association membership, and positive livelihood perceptions compared to non-beneficiaries. Furthermore, propensity score matching indicates a statistically significant positive association between program participation and coded total current income levels.

Nevertheless, because covariate balance was not fully achieved within the specified threshold and the evaluation relied on a cross-sectional no-baseline design, the results must be interpreted with restraint. The findings provide descriptive evidence of positive alignment between program receipt and socioeconomic indicators among fisherfolk.

4.3 RECOMMENDATIONS

1. Expand the distribution of essential processing and packaging equipment, such as vacuum packers and solar dryers, to address the remaining gaps where substantial percentages of beneficiaries still report lacking equipment.
2. Strengthen support for organized fisherfolk associations, given that membership strongly predicts program participation and correlates with active livelihood engagement.

3. Enhance technical training programs focusing on fish processing methods to further increase the proportion of beneficiaries actively engaging in value-adding activities like drying and deboning.
4. Target outreach to non-beneficiary communities to increase general program awareness and extend livelihood support to eligible households currently outside the program.
5. Incorporate baseline data collection in future iterations of program evaluations to strengthen causal inference and bypass the limitations of cross-sectional matching designs.
6. Refine matching protocols and sample selection in subsequent evaluations to ensure that all covariate balance thresholds are fully satisfied.